

CyberGuard AI - Final Report

AI-Powered Home Network Security Monitor for IoT Devices

GitHub: <https://github.com/AbhiramKallepalli/cyberguard-ai>

1. Business Problem and Motivation

The rapid adoption of Internet of Things (IoT) devices in homes has introduced significant security risks that most homeowners are unprepared to handle. Smart TVs, security cameras, thermostats, smart bulbs, voice assistants, and home NAS servers are now common household devices, and each one represents a potential entry point for malicious actors.

Unlike corporate environments, where dedicated Security Operations Center (SOC) teams monitor network activity around the clock using enterprise-grade tools, homeowners have no equivalent protection. Commercial security platforms such as Darktrace and CrowdStrike are designed for enterprise networks and cost thousands of dollars per month, well beyond the reach of a typical household.

The specific problem this project addresses is: how can a non-technical homeowner be alerted when their IoT devices are behaving suspiciously, and understand what to do about it, without needing any technical expertise?

This problem is widespread, since tens of millions of households now own multiple IoT devices; it is underserved, since no affordable, user-friendly AI security tool exists for home networks; and it is high-impact, since compromised IoT devices can expose home video footage, banking sessions, and personal data.

The target user is a non-technical homeowner who owns IoT devices such as a smart TV, security camera, thermostat, smart bulbs, or a home NAS, but who lacks technical knowledge, enterprise security tools, or the budget for commercial solutions. What this user needs is simple, plain-English alerts and clear guidance on what action to take.

CyberGuard AI exists to help a non-technical homeowner detect unusual activity on their home network, understand what it means in plain English, and decide what action to take, by combining an unsupervised anomaly detection model with a local LLM-powered assistant.

2. Architecture and Design Decisions

The system follows a three-layer architecture, separating detection, explanation, and interpretation into distinct, disciplined components.

Layer 1 - Anomaly Detection Engine (Isolation Forest). This layer handles all anomaly detection tasks. It is trained on the NSL-KDD network intrusion dataset and flags network activity that deviates significantly from learned normal patterns. This layer is fully deterministic and does not involve the LLM. It uses an unsupervised approach, meaning the model learns what normal traffic looks like without ever being told what attacks look like in advance. This mirrors real-world conditions, since no homeowner has a dataset of their own traffic pre-labeled with attacks.

Layer 2 - Explainability Layer (SHAP). This layer explains why each flagged event was considered anomalous by identifying the specific features that contributed most to the anomaly score. To keep the system responsive, SHAP is run only on the top 500 highest-risk events per scan rather than the full dataset, and outputs are translated into plain-English descriptions before being shown to the homeowner (for example, “unusually large amount of data received” rather than a raw feature name like “src_bytes”).

Layer 3 - LLM Assistant (Local LLM via Ollama). This layer handles all user interaction. The homeowner can ask free-form questions about any flagged event, and the LLM receives only structured outputs from Layers 1 and 2: the anomaly score, risk category, and top contributing features. It does not have access to raw network data, and it does not make decisions; it only interprets and explains. This disciplined separation between prediction and explanation mirrors the same pattern used successfully in Project 1 (the AI Decision Copilot for Customer Retention), where XGBoost made predictions and a local LLM handled only Q&A.

Why Isolation Forest over supervised classification? Supervised models such as XGBoost, used in Project 1, require labeled training data explicitly marked as normal or attack. In the real world, most home network traffic is unlabeled. Isolation Forest is an unsupervised anomaly detection algorithm that learns what normal traffic looks like and flags deviations without ever being shown labeled attack examples, making it a more realistic and differentiated choice for this problem.

Why a local LLM over a cloud API? Home network logs contain sensitive information about device activity and usage patterns. Sending this data to a cloud API would create a privacy risk. A local LLM running via Ollama processes all data on the homeowner's own machine, with no data leaving the device at any point.

The data flow is as follows: the homeowner uploads a CSV network log file; the Flask backend validates the file; the Isolation Forest model generates anomaly scores; the SHAP layer identifies top contributing features per flagged event; a plain-English translation layer converts technical feature names into homeowner-friendly language; results are stored in a SQL database and displayed on the results page; and finally, the homeowner can ask a question about a flagged event, with the LLM responding using only the structured context passed to it.

The technology stack consists of Flask as the backend framework, SQLite with SQLAlchemy for the database, Isolation Forest for anomaly detection, SHAP for explainability, Ollama with Phi-3 Mini for the local LLM, HTML with Jinja2 templating for pages, custom CSS for styling, JavaScript and Chart.js for interactivity, the NSL-KDD dataset for training, and pytest for automated testing.

3. Implementation Details

The database uses four relational tables. The users table stores account information with hashed passwords. The scans table records each uploaded file, linked to a user, with total event and flagged event counts. The flagged_events table stores each anomaly's score, risk category, plain-English description, and JSON-encoded top SHAP factors, linked to a scan. The llm_responses table stores each homeowner question and the LLM's answer, linked to a flagged event.

The Flask application exposes eight routes: the dashboard, login, signup, logout, upload, scan results, event detail, and an AJAX endpoint for asking the LLM a question. Authentication is handled with Flask-Login and Werkzeug password hashing, and every route except login and signup requires an authenticated session.

The preprocessing pipeline, implemented in preprocessing.py, reads an uploaded CSV entirely in memory rather than persisting it to disk, in line with the project's privacy requirements. It validates that the file has between 41 and 43 columns matching the NSL-KDD schema, handles missing values, encodes the three categorical columns (protocol type, service, and flag) using label encoding, and scales the remaining

numeric columns using standardization. Invalid files, including empty files, malformed CSVs, and files with an incorrect column count, are rejected with a clear, specific error message rather than allowing the application to crash or silently produce incorrect results.

The model layer, implemented in `model.py`, trains an Isolation Forest on the NSL-KDD training set without using its attack labels, then applies it to uploaded data to generate a continuous anomaly score for each record. Records are classified into LOW, MEDIUM, or HIGH risk categories based on score thresholds. SHAP's explainability analysis is run on the top 500 highest-risk events per scan to keep processing time reasonable on consumer hardware, and each event's top three contributing factors are translated into a plain-English dictionary mapping technical NSL-KDD feature names to homeowner-friendly descriptions.

The frontend polish completed in Week 8 included an AJAX-based Q&A interface so homeowners can ask questions about a flagged event without a full page reload, a five-stage live progress indicator during file upload and scanning, a Chart.js bar chart on the dashboard summarizing alerts per scan, and a tightened LLM prompt that constrains Phi-3 Mini's responses to three clean sentences grounded strictly in the event's structured data.

4. Model Evaluation Results

The Isolation Forest model was trained on the NSL-KDD training set (125,973 records, 41 features) without using the dataset's normal/attack labels, consistent with the unsupervised approach described in Section 2. The model was then evaluated by applying it to network logs and comparing its flagging behavior against known outcomes through structured testing.

In Scenario 2 testing, a sample of 50 records labeled as attacks in the NSL-KDD test set was uploaded. The model flagged 39 of the 50 records (78%) as high-risk, and the SHAP explainability layer surfaced factors consistent with genuine attack behavior, including unusually large data volumes, high connection error rates, and multiple compromised conditions detected. This result validates that the model, despite never being trained on attack labels, successfully learns to distinguish attack-like patterns from the bulk of normal training data.

In Scenario 1 testing, a sample of 50 records labeled as normal traffic was uploaded, with the expectation (per the original specification) that none of these would be flagged. In practice, all 50 records (100%) were flagged as high risk. This is discussed in detail in Section 6 (Limitations), but the key finding is that Isolation Forest's unsupervised nature means it flags statistical outliers relative to the training distribution as a whole, not “attacks” specifically, and a small sample of normal traffic can still appear anomalous relative to that broader distribution.

To satisfy the spec's evaluation requirement (Section 9.1), the model was formally evaluated against the full NSL-KDD test set (22,544 records) using its ground-truth normal/attack labels, which were withheld during training and used here only for post-hoc validation. Against these labels, the model achieved a precision of 0.838, a recall of 0.686, and an F1-score of 0.755. In other words, when the model flags a record as anomalous, it is correct 83.8% of the time, and it successfully identifies 68.6% of all true attack records in the test set. The anomaly score distributions also confirm meaningful separation between the two classes: normal records averaged a score of -0.477 (median -0.476, $n=9,711$), while attack records averaged a more negative -0.530 (median -0.552, $n=12,833$), consistent with Isolation Forest's convention that more negative scores indicate greater anomalousness. This separation, achieved entirely without the model ever seeing attack labels during training, validates the core premise of the unsupervised approach described in Section 2.

Across all testing, the anomaly scores produced showed meaningful separation and variation rather than collapsing to a single value, and the model consistently produced a valid classification and score for every record processed, with no crashes or invalid outputs observed during testing.

During final review, an additional issue was identified and corrected: the risk category thresholds (LOW/MEDIUM/HIGH) had originally been set to fixed values (-0.15 and -0.05) that did not match the actual range of scores produced by this model on NSL-KDD data, which falls between approximately -0.70 and -0.40. As a result, all 22,544 records in the test set were being classified as HIGH risk regardless of their true relative anomaly level. This was diagnosed by analyzing the full score distribution on the test set and recalibrating the thresholds to the 33rd and 66th percentiles of that distribution (-0.5453 and -0.4717), producing a realistic three-way split of approximately 33% HIGH, 33% MEDIUM, and 34% LOW across the test set. All 16 automated tests were re-run and confirmed passing after this change, and Scenario 2 was re-executed, producing a mix of 34 HIGH and 5 MEDIUM classifications among the 39 flagged attack records, consistent with expectations.

5. LLM Evaluation Results

Five LLM responses were generated using Phi-3 Mini via Ollama, each grounded in a real flagged event's anomaly score, risk category, and SHAP-derived top factors, and scored against the project's 0–3 rubric (3 = directly answers, cites specific anomaly data, no fabrication, plain language; 2 = answers but is partially vague or technical; 1 = generic or only partially answers; 0 = incorrect or hallucinated).

Four of the five responses scored a 3. These responses consistently referenced the specific risk level, anomaly score, and named SHAP factors for the event in question (for example, “unusually large amount of data received,” “high REJ errors at destination,” “multiple compromised conditions detected”), used plain non-technical language, and recommended a concrete next action such as disconnecting the device or running an antivirus scan. One response, when asked to identify the single most concerning factor, correctly isolated the most relevant SHAP factor and offered an appropriately hedged technical interpretation (a possible DDoS pattern or misconfiguration) while still recommending professional follow-up rather than overstating certainty.

One response scored a 2. When asked what a flagged event meant “in simple terms,” the LLM correctly referenced the risk level and anomaly score but described the underlying cause generically as “unusual activity” rather than citing the event's specific SHAP factors (high connections to a destination host, repeated use of the same source port, and connections to many services). The response was accurate and non-fabricated, but less precisely grounded than the other four.

The average score across all five responses was 2.8 out of 3.0. No response introduced information outside the structured context passed to it, confirming that the prompt design, which restricts the LLM to interpreting rather than predicting, successfully prevents hallucination even when the underlying event data is complex.

6. Limitations and Future Work

The most significant limitation surfaced during testing concerns Scenario 1. The project specification's expected outcome was that a normal-traffic-only file would produce no flagged events. In testing, all 50 sampled normal records were flagged as HIGH risk. This is a genuine limitation of the unsupervised approach as currently tuned, not a bug or crash: Isolation Forest scores every record relative to the statistical distribution of its training data, which itself contains a mix of normal and attack traffic, rather than relative to a clean baseline of only normal traffic. As a result, a small sample of normal records can still register as

statistical outliers if the risk thresholds are tuned aggressively. Future work should explore retraining or threshold calibration using only normal-labeled NSL-KDD records as the baseline distribution, which may better align the model's flagging behavior with homeowner expectations of “quiet” periods producing few or no alerts.

A second limitation is dataset realism. NSL-KDD is a widely used academic benchmark for network intrusion detection, but it was not collected from real consumer IoT devices, and its 41 features (protocol type, byte counts, connection statistics, and so on) describe generic network connections rather than IoT-specific behaviors such as device-to-cloud telemetry patterns or smart-home protocol traffic. A production version of this tool would benefit from either a purpose-built IoT traffic dataset or a transfer-learning approach that fine-tunes on real home network captures.

A third limitation is scale. SHAP explainability is currently run only on the top 500 highest-risk events per scan to keep processing time practical on consumer hardware; on the 125,973-record training file, this means the vast majority of events never receive an explanation, even if some among them are meaningfully anomalous. A future version could use a faster approximate SHAP method or a smaller, pre-filtered candidate set to explain a larger share of events without a prohibitive runtime cost.

Finally, this project intentionally scoped several features out of Version 1 in favor of a stable, well-tested core: per-device dashboards and activity trends, device history tracking across multiple scans, case management (marking events as checked, safe, or confirmed problems, with notes), scan history comparison, and actionable LLM recommendations with specific remediation steps such as changing a device password. These are documented as Version 2 features in the original specification and represent the natural next phase of development once V1 is fully validated.

Despite these limitations, the core system meets its primary goal: a non-technical homeowner can upload a network log, receive plain-English, color-coded risk assessments, and ask free-form follow-up questions that are answered accurately and without fabrication, entirely offline and without any data leaving their own machine.

7. Conclusion

CyberGuard AI demonstrates a complete, disciplined three-layer architecture for home network anomaly detection: an unsupervised Isolation Forest model for detection, SHAP for explainability, and a local LLM strictly confined to interpretation rather than prediction. All six V1 features specified at project kickoff were implemented, tested, and verified working end to end. Automated testing (16 pytest unit tests across model, preprocessing, and database logic) and all six specification test scenarios were executed and documented, with five of six scenarios matching expected behavior and the sixth yielding a well-understood, documented limitation rather than an unexplained failure. Formal evaluation against the full labeled test set produced a precision of 0.838, recall of 0.686, and F1-score of 0.755, confirming the unsupervised model meaningfully distinguishes attack from normal traffic despite never seeing attack labels during training. LLM response quality averaged 2.8 out of 3.0 across five grounded evaluations, with zero fabricated responses observed. The project is complete, fully committed to GitHub, and ready for review.